

# Deliverable 5 – KPI Definition Document

## Healthcare Patient Experience Analytics using Power BI

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### Introduction

Key Performance Indicators (KPIs) are measurable metrics used to evaluate healthcare performance and patient satisfaction. In this project, KPIs were developed using **Data Analysis Expressions (DAX)** in Microsoft Power BI to provide meaningful insights into national, regional, and state-level healthcare performance.

These KPIs enable healthcare stakeholders to monitor patient satisfaction, compare healthcare quality across different regions, identify performance trends, benchmark states against national averages, and support evidence-based decision-making.

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### KPI Definitions

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#### 1. National Top-box %

##### Definition

Calculates the average percentage of patients who selected the most positive response (Top-box) across all national survey results.

##### DAX Formula

```
National Top-box % =  
CALCULATE(  
    AVERAGE('National Results'[Percentage]),  
    Responses[Response] = "Top-box"  
)
```

##### Business Importance

- Measures overall national patient satisfaction.
- Serves as a benchmark for state-level comparisons.

- Supports executive-level healthcare performance monitoring.
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## 2. National Middle-box %

### Definition

Calculates the average percentage of patients selecting the Middle-box response category.

### DAX Formula

```
National Middle-box % =  
CALCULATE(  
    AVERAGE('National Results'[Percentage]),  
    Responses[Response] = "Middle-box"  
)
```

### Business Importance

- Indicates moderate patient satisfaction.
  - Helps identify opportunities for improving patient experience.
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## 3. National Bottom-box %

### Definition

Calculates the average percentage of patients selecting the least positive response category.

### DAX Formula

```
National Bottom-box % =  
CALCULATE(  
    AVERAGE('National Results'[Percentage]),  
    Responses[Response] = "Bottom-box"  
)
```

### Business Importance

- Identifies dissatisfaction among patients.
  - Highlights areas requiring immediate improvement.
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## 4. State Average Top-box %

### Definition

Calculates the average Top-box percentage for the selected state.

## DAX Formula

```
State Average Top-box % =  
CALCULATE(  
    AVERAGE('State Results'[Percentage]),  
    Responses[Response] = "Top-box"  
)
```

## Business Importance

- Evaluates patient satisfaction at the state level.
  - Supports state performance comparisons.
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# 5. State Satisfaction Score

## Definition

Represents the average patient satisfaction score for the selected state.

## DAX Formula

```
State Satisfaction Score =  
AVERAGE('State Results'[Percentage])
```

## Business Importance

- Measures overall healthcare quality within a state.
  - Helps identify high-performing and underperforming states.
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# 6. Regional Satisfaction Score

## Definition

Calculates the average patient satisfaction score for the selected region.

## DAX Formula

```
Regional Satisfaction Score =  
AVERAGE('State Results'[Percentage])
```

## Business Importance

- Compares healthcare quality across different regions.
  - Supports regional healthcare planning.
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## 7. Survey Response Rate

### Definition

Measures the average survey response percentage.

### DAX Formula

```
Survey Response Rate =  
AVERAGE(Responses[Percentage])
```

### Business Importance

- Indicates participation in healthcare surveys.
  - Helps assess the reliability of survey-based insights.
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## 8. Number of Participating States

### Definition

Counts the total number of states included in the analysis.

### DAX Formula

```
Number of Participating States =  
DISTINCTCOUNT(States[State Code])
```

### Business Importance

- Indicates survey coverage.
  - Helps evaluate dataset completeness.
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## 9. Number of Quality Measures

### Definition

Counts the healthcare quality measures evaluated in the survey.

### DAX Formula

```
Number of Quality Measures =  
DISTINCTCOUNT(Measures[Measure ID])
```

### Business Importance

- Shows the breadth of healthcare performance evaluation.
- Supports dashboard summaries.

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## 10. Highest Performing Measure

### Definition

Identifies the healthcare quality measure with the highest patient satisfaction.

### DAX Formula

```
Highest Performing Measure =  
TOPN(  
    1,  
    VALUES(Measures[Measure Name]),  
    [State Satisfaction Score],  
    DESC  
)
```

### Business Importance

- Identifies healthcare strengths.
  - Highlights best-performing service areas.
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## 11. Lowest Performing Measure

### Definition

Identifies the healthcare quality measure with the lowest patient satisfaction.

### DAX Formula

```
Lowest Performing Measure =  
TOPN(  
    1,  
    VALUES(Measures[Measure Name]),  
    [State Satisfaction Score],  
    ASC  
)
```

### Business Importance

- Highlights improvement opportunities.
  - Supports quality improvement initiatives.
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## 12. Year-over-Year Satisfaction Change

### Definition

Calculates the change in patient satisfaction between reporting periods.

### DAX Formula

```
Year-over-Year Satisfaction Change =  
[Current Year Satisfaction] -  
[Previous Year Satisfaction]
```

### Business Importance

- Measures improvement over time.
  - Supports long-term performance monitoring.
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## 13. Regional Ranking

### Definition

Ranks regions based on average patient satisfaction.

### DAX Formula

```
Regional Ranking =  
RANKX(  
    ALL(States[Region]),  
    [Regional Satisfaction Score],  
    ,  
    DESC,  
    DENSE  
)
```

### Business Importance

- Identifies the highest-performing healthcare regions.
  - Supports regional benchmarking.
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## 14. State Ranking

### Definition

Ranks states according to patient satisfaction scores.

## DAX Formula

```
State Ranking =  
RANKX(  
    ALL(States[State Name]),  
    [State Satisfaction Score],  
    ,  
    DESC,  
    DENSE  
)
```

## Business Importance

- Compares state healthcare performance.
  - Supports benchmarking and performance monitoring.
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## KPI Summary

| KPI                                | Purpose                                 |
|------------------------------------|-----------------------------------------|
| National Top-box %                 | National patient satisfaction benchmark |
| National Middle-box %              | Moderate satisfaction analysis          |
| National Bottom-box %              | Patient dissatisfaction analysis        |
| State Average Top-box %            | State-level satisfaction                |
| State Satisfaction Score           | Overall state performance               |
| Regional Satisfaction Score        | Regional healthcare performance         |
| Survey Response Rate               | Survey participation analysis           |
| Number of Participating States     | Dataset coverage                        |
| Number of Quality Measures         | Survey scope                            |
| Highest Performing Measure         | Best-performing healthcare service      |
| Lowest Performing Measure          | Improvement opportunity                 |
| Year-over-Year Satisfaction Change | Trend analysis                          |
| Regional Ranking                   | Regional comparison                     |
| State Ranking                      | State comparison                        |

## Conclusion

The KPIs developed in this project provide a comprehensive framework for evaluating patient satisfaction across national, regional, and state levels. By combining DAX calculations with interactive Power BI dashboards, these metrics enable healthcare stakeholders to monitor

performance, identify improvement opportunities, benchmark results, and make informed decisions that support continuous healthcare quality improvement.

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